

DTU



Perception for Autonomous Systems 31392:

Fitting

Lecturer: Evangelos Boukas—PhD

Fitting Data to a Model (Handling Outliers)

- Let's work with the line example
 - Fitting a model without (or with minimum) outliers data points
 - Fitting data through voting (Hough Transform)
 - RANdom SAmple Consensus (RANSAC)
- What about data not in a line?

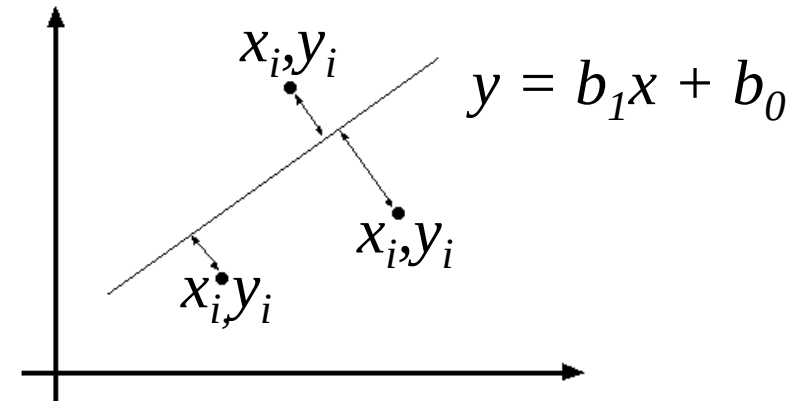
Least Squares

- Let the line depicted here be described by:
 $y = b_1x + b_0$
- Then the distance of a point x_i, y_i is defined as:

$$|ax_i + by_i + c|$$

- Therefore, we can find the line that best matches our data by minimizing the following function:

$$E = \sum_{i=1}^n (ax_i + by_i + c)^2$$



Least Squares

- To form it as a least squares problem we need to minimize:

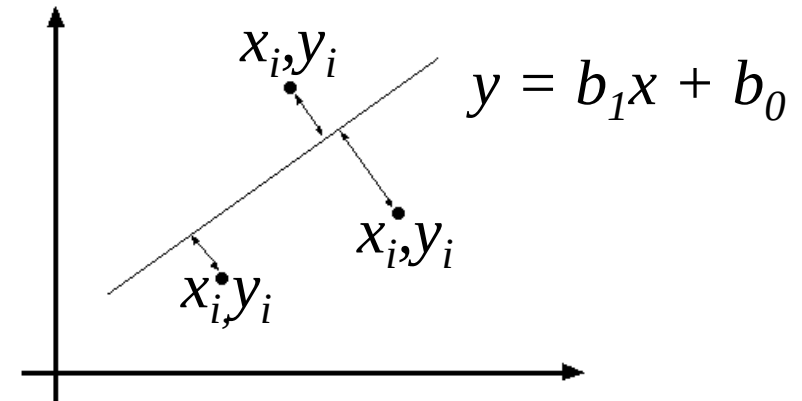
$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

where y_i, x_i are the observed values, and β_0, β_1 are the unknown parameters to be estimated.

- The closed-form solution for the slope (β_1) and intercept (β_0) :

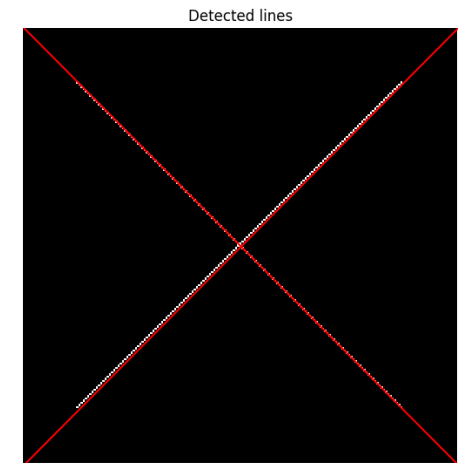
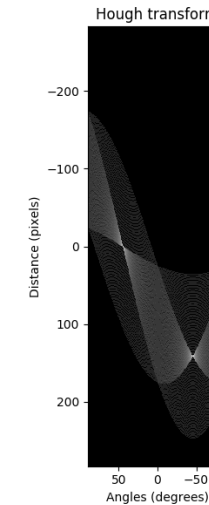
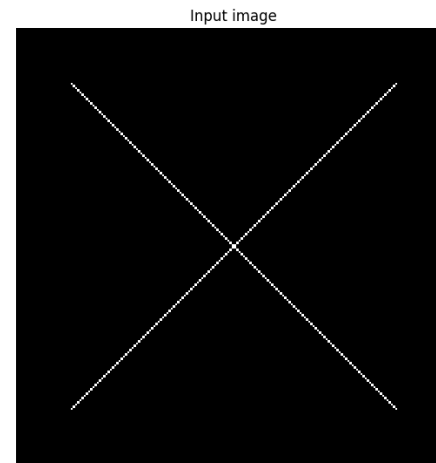
$$\beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \beta_0 = \bar{y} - \beta_1 \bar{x}$$

- Finally, the predicted line equation: $\hat{y} = \beta_0 + \beta_1 x$
- However, for multiple regressions the problem is a normal equation one, requiring matrix inversion. With large data this problem is not directly solvable (in a reasonable time in a closed form solution)



Hough Transform

- Assuming we want to fit a line in our data we can use the Hough transform as follows:
 - Formulate the problem as a bounded one
 - Create a grid of parameter values
 - Each data point votes on the grid
 - Find *local*-maxima in the grid and track back to lines in image



Hough Transform (1/4)

- Let's consider the line equation:

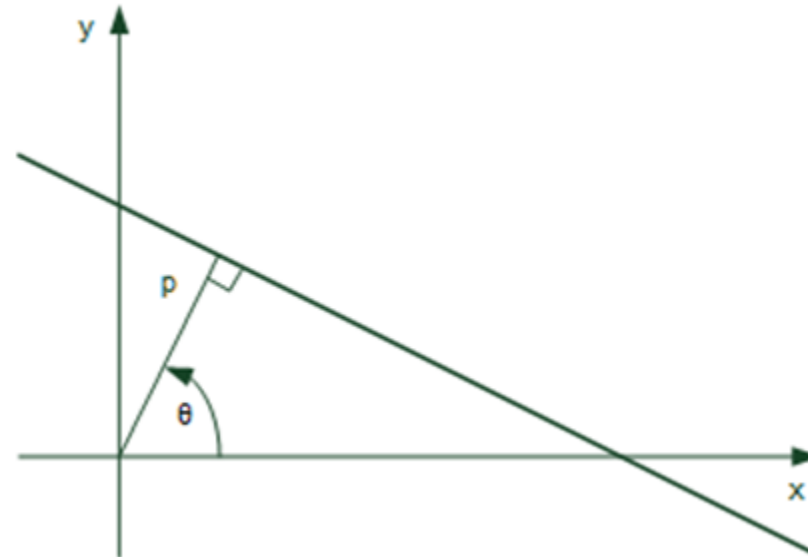
$$Y=mX+b$$

- The problem with the above equation is that b, m are unbounded, Therefore, we consider the following formulation (polar transformation):

$$x \cos \theta - y \sin \theta = \rho$$

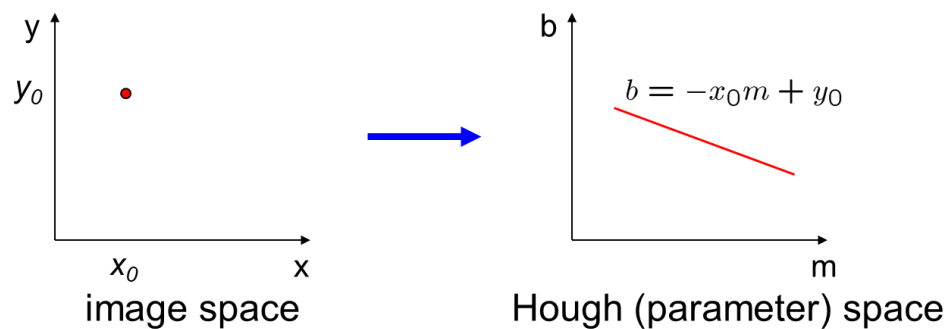
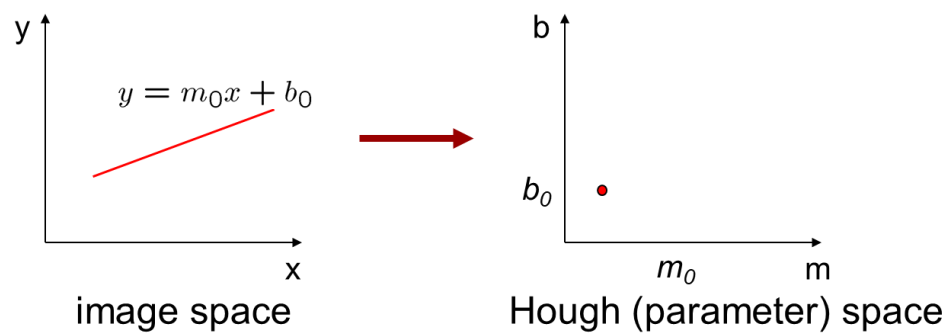
ρ : perpendicular distance from line to origin

θ : angle the perpendicular makes with the x-axis

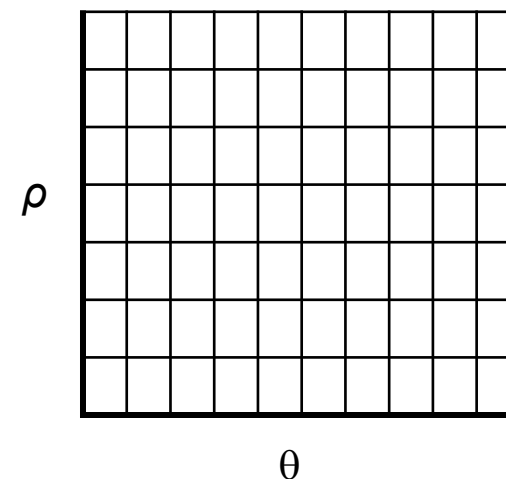


Hough Transform (2/4)

- Next step:
 - Initialize the grid using (d, θ)

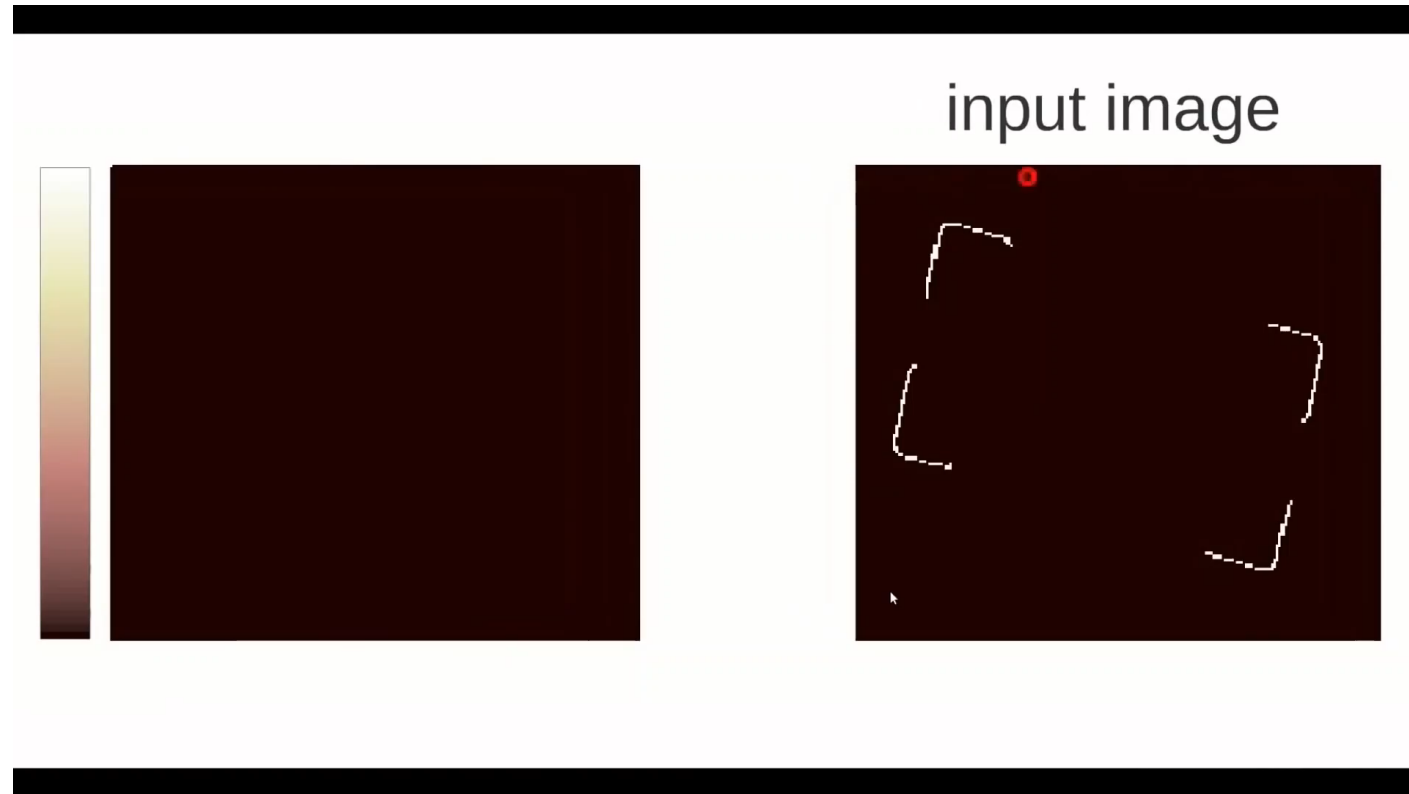


H: accumulator array (votes)



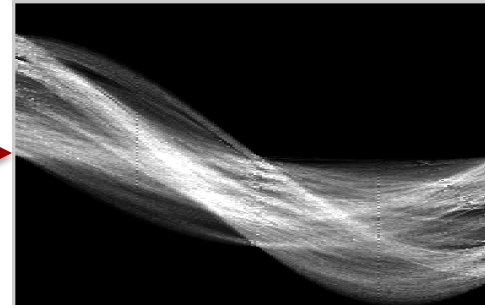
Hough Transform (3/4)

- Populate the grid by passing through the whole image and adding votes.
- See the following video:



Hough Transform (4/4)

- Identify maxima and track lines back to image:



What about data not in a line?

- Same approach is followed for more complex models
eg: circle model is:

$$(x - x_0)^2 + (y - y_0)^2 = r^2$$

which requires 3 parameters in the Hough grid

- However, the complexity grows exponentially
(usually up to 4 parameters is advised)
- Ransac can handle higher order models.



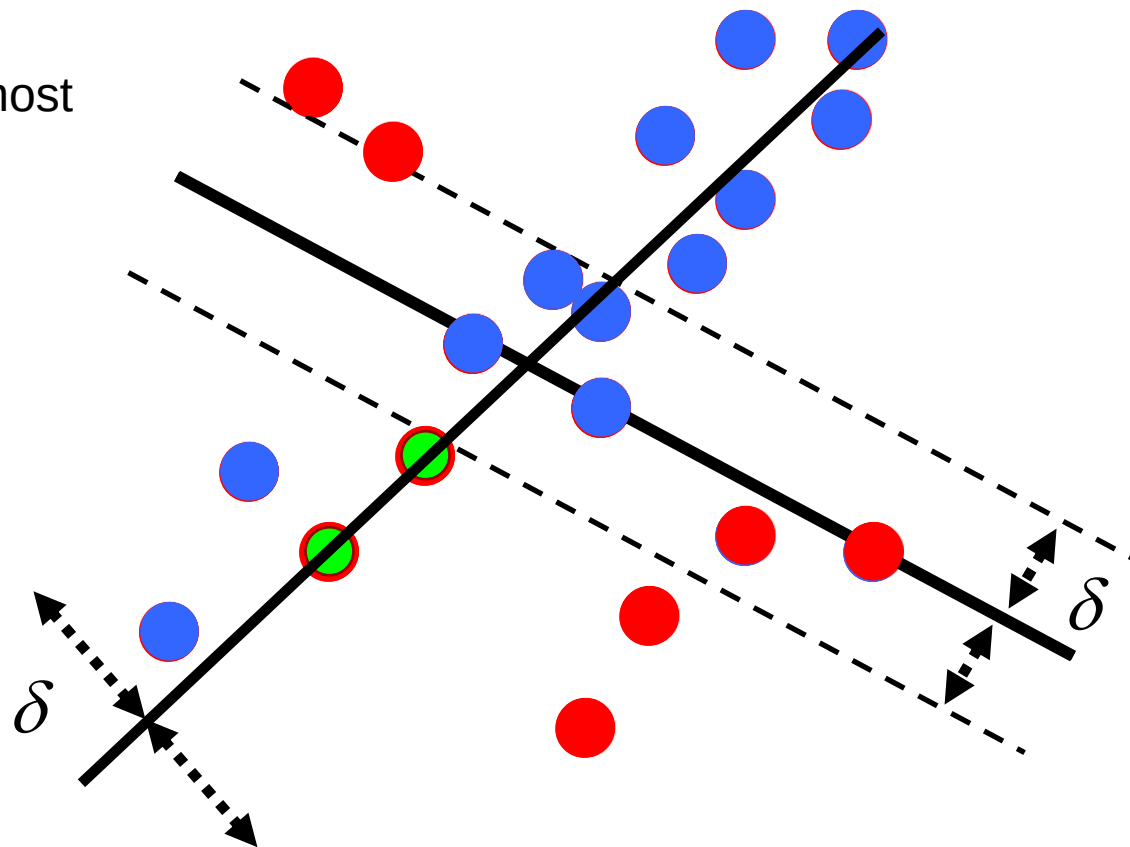
RANdom SAmple Consensus (RANSAC)

Algorithm:

1. **Sample** (randomly) the number of points required to fit the model
 2. **Solve** for model parameters using samples
 3. **Score** by the fraction of inliers within a preset threshold of the model
- **Repeat** 1-3 until the best model is found with high confidence
 - δ - is the threshold upon which a sample is considered to not fit to the selected model

RANdom SAmple Consensus (RANSAC)

- Select the models with most inliers to create lines



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